

July 2025 CSE 220

Online on CFT

Subsection: A1+A2

Time: 30 minutes

Consider the continuous-time signal:

$$x(t) = 0.5\cos(4t) + 0.5\sin(6t)$$

Now, we define

$$y_1(t) = \frac{d}{dt}x(t)$$

Compute the Continuous Fourier Transform for: $x(t)$, $y_1(t)$, and $y_2(t)$

The theoretical CFT property is:

$$\mathcal{F}\left\{\frac{d}{dt}x(t)\right\} = j2\pi fX(f)$$

1. Compute:

$$j2\pi fX(f)$$

2. Derive the expression for $\frac{d}{dt}x(t)$ and find CFT of that function. $Y_1(f) = \mathcal{F}\left\{\frac{d}{dt}x(t)\right\}$.

2. Plot and verify overlap:

- $|Y_1(f)|$ vs $|j2\pi fX(f)|$
- phase comparison

3. Repeat this procedure for 2nd and 3rd derivative.

4. Comment on:

- MSE analysis for both magnitude and phase for all 3 derivatives..

Implementation Constraints

- You are NOT allowed to use `np.fft` or any built-in Fourier transform function.
- You must use `np.trapz()` or `np.trapezoid()` for numerical integration.
- Submit a single Python file named: `StudentID.py`